

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

07

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Q1**A****A. -6**

Choice A is correct. It's given that the graph of $y = fx$ in the xy -plane passes through the points $7, 0$ and $-3, 0$. It follows that when the value of x is either 7 or -3, the value of fx is 0. It's also given that the function f is defined by $fx = ax^2 + bx + c$, where a , b , and c are constants. It follows that the function f is a quadratic function and, therefore, may be written in factored form as $fx = ax - ux - v$, where the value of fx is 0 when x is either u or v . Since the value of fx is 0 when the value of x is either 7 or -3, and the value of fx is 0 when the value of x is either u or v , it follows that u and v are equal to 7 and -3. Substituting 7 for u and -3 for v in the equation $fx = ax - ux - v$ yields $fx = ax - 7x - 3$, or $fx = ax - 7x + 3$. Distributing the right-hand side of this equation yields $fx = ax^2 - 7x + 3x - 21$, or $fx = ax^2 - 4ax - 21a$. Since it's given that $fx = ax^2 + bx + c$, it follows that $b = -4a$. Adding a to each side of this equation yields $a + b = -3a$. Since $a + b = -3a$, if a is an integer, the value of $a + b$ must be a multiple of 3. If a is an integer greater than 1, it follows that $a \geq 2$. Therefore, $-3a \leq -6$. It follows that the value of $a + b$ is less than or equal to -6. Of the given choices, only -6 is a multiple of 3 that's less than or equal to -6.

Choice B is incorrect. This is the value of $a + b$ if a is equal to, not greater than, 1.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**C**

C. The unit price, in dollars, of the product that will result in a revenue of \$0

Choice C is correct. By definition, the y-value when a function intersects the x-axis is 0. It's given that the graph of the function intersects the x-axis at 0 and a, that x is the unit price, in dollars, of a product, and that $f(x)$, where $y = f(x)$, is the revenue, in dollars, that a company receives from the sales of the product. Since the value of a occurs when $y = 0$, a is the unit price, in dollars, of the product that will result in a revenue of \$0.

Choice A is incorrect. The revenue, in dollars, when the unit price of the product is \$0 is represented by $f(x)$, when $x = 0$. Choice B is incorrect. The unit price, in dollars, of the product that will result in maximum revenue is represented by the x-coordinate of the maximum of f. Choice D is incorrect. The maximum revenue, in dollars, that the company can make is represented by the y-coordinate of the maximum of f.

Q3**B**

B. $-1 < r < 0$

Choice B is correct. The term $(1 + r)$ represents a percent change. Since the population is decreasing, the percent change must be between 0% and 100%. When the percent change is expressed as a decimal rather than as a percent, the percentage change must be between 0 and 1. Because $(1 + r)$ represents percent change, this can be expressed as $0 < 1 + r < 1$. Subtracting 1 from all three terms of this compound inequality results in $-1 < r < 0$.

Choice A is incorrect. If $r < -1$, then after 1 year, the population P would be a negative value, which is not possible. Choices C and D are incorrect. For any value of $r > 0$, $1 + r > 1$, and the exponential function models growth for positive values of the exponent. This contradicts the given information that the population is decreasing.

Q4**210**

The correct answer is 210. Since -5 , 6, and 7 are zeros of the function, the function can be rewritten as $h(x) = (x+5)(x-6)(x-7)$. Expanding the function yields $h(x) = x^3 - 8x^2 - 23x + 210$. Thus, $a = -8$, $b = -23$, and $c = 210$. Therefore, the value of c is 210.

Q5**5**

The correct answer is 5. It's given that $fx = -a^x + b$. Substituting $-a^x + b$ for fx in the equation $y = fx - 15$ yields $y = -a^x + b - 15$. It's given that the y -intercept of the graph of $y = fx - 15$ is 0, $-\frac{99}{7}$. Substituting 0 for x and $-\frac{99}{7}$ for y in the equation $y = -a^x + b - 15$ yields $-\frac{99}{7} = -a^0 + b - 15$, which is equivalent to $-\frac{99}{7} = -1 + b - 15$, or $-\frac{99}{7} = b - 16$. Adding 16 to both sides of this equation yields $\frac{13}{7} = b$. It's given that the product of a and b is $\frac{65}{7}$, or $ab = \frac{65}{7}$. Substituting $\frac{13}{7}$ for b in this equation yields $a\frac{13}{7} = \frac{65}{7}$. Dividing both sides of this equation by $\frac{13}{7}$ yields $a = 5$.

Q6**D****D. Neither I nor II**

Choice D is correct. A function defined by an equation in the form $fx = ab^{x+h}$, where a , b , and h are positive constants and $x \geq 0$, has a minimum value of $f(0)$. It's given that function g is defined by $gx = 181.161.4^{x+2}$, which is equivalent to $gx = 20.881.4^{x+2}$. Substituting 0 for x in this equation yields $g0 = 20.88(1.4)^{0+2}$, or $g(0) = 40.9248$. Therefore, the minimum value of $g(x)$ is 40.9248, so $gx = 181.161.4^{x+2}$ doesn't display its minimum value as a constant or coefficient. It's also given that function h is defined by $hx = 181.4^{x+4}$. Substituting 0 for x in this equation yields $h0 = 181.4^{0+4}$, or $h(0) = 69.1488$. Therefore, the minimum value of $h(x)$ is 69.1488, so $hx = 181.4^{x+4}$ doesn't display its minimum value as a constant or coefficient. Therefore, neither I nor II displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7**A****A.** -4

Choice A is correct. The given quadratic function f is in vertex form, $f(x) = (x - h)^2 + k$, where (h, k) is the vertex of the graph of $y = f(x)$ in the xy -plane. Therefore, the vertex of the graph of $y = f(x)$ is $(2, -4)$. In addition, the y -coordinate of the vertex represents either the minimum or maximum value of a quadratic function, depending on whether the graph of the function opens upward or downward. Since the leading coefficient of f (the coefficient of the term $(x - 2)^2$) is 1, which is positive, the graph of $y = f(x)$ opens upward. It follows that at $x = 2$, the minimum value of the function f is -4 .

Choice B is incorrect and may result from making a sign error and from using the x -coordinate of the vertex. Choice C is incorrect and may result from using the x -coordinate of the vertex. Choice D is incorrect and may result from making a sign error.

Q8**D****D.** $y = 2004^x$

Choice D is correct. Since the value of y increases by a constant factor, 4, for each increase of 1 in the value of x , the relationship between x and y is exponential. An exponential relationship between x and y can be represented by an equation of the form $y = ab^x$, where a is the value of y when $x = 0$ and y increases by a factor of b for each increase of 1 in the value of x . Since $y = 200$ when $x = 0$, $a = 200$. Since y increases by a factor of 4 for each increase of 1 in the value of x , $b = 4$. Substituting 200 for a and 4 for b in the equation $y = ab^x$ yields $y = 2004^x$. Thus, the equation $y = 2004^x$ represents the relationship between x and y .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This equation represents a relationship where for each increase of 1 in the value of x , the value of y increases by a factor of 200, not 4, and when $x = 0$, y is equal to 4, not 200.

Choice C is incorrect and may result from conceptual errors.

Q9**D**

- D. The estimated number of advertisements the company sent to its clients in 1997 was 9,000.
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Choice D is correct. The y -intercept of a graph in the xy -plane is the point where $x = 0$. For the given function f , the y -intercept of the graph of $y = fx$ in the xy -plane can be found by substituting 0 for x in the equation $y = 9,000(0.66)^x$, which gives $y = 9,000(0.66)^0$. This is equivalent to $y = 9,000(1)$, or $y = 9,000$. Therefore, the y -intercept of the graph of $y = fx$ is 0, 9,000. It's given that the function f models the number of advertisements a company sent to its clients each year. Therefore, fx represents the estimated number of advertisements the company sent to its clients each year. It's also given that x represents the number of years since 1997. Therefore, $x = 0$ represents 0 years since 1997, or 1997. Thus, the best interpretation of the y -intercept of the graph of $y = fx$ is that the estimated number of advertisements the company sent to its clients in 1997 was 9,000.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10

A

A. 23

Choice A is correct. It's given that in the xy -plane, the graph of $y = g(x)$ passes through the point $(0, 22)$. It follows that $g(0) = 22$. Substituting 0 for x and 22 for $g(x)$ in the given equation yields $22 = \frac{0^2 - 0 - a}{0^3 - 0 - b}$, which is equivalent to $22 = \frac{-a}{-b}$, or $22 = \frac{a}{b}$. Multiplying each side of this equation by b yields $22b = a$. Substituting $22b$ for a in the equation defining the function g yields $gx = \frac{x^2 - x - 22b}{x^3 - x - b}$. It's given that $g(-22) = 0$. Substituting -22 for x and 0 for $g(x)$ in the equation $gx = \frac{x^2 - x - 22b}{x^3 - x - b}$ yields $0 = \frac{(-22)^2 - (-22) - 22b}{(-22)^3 - (-22) - b}$. Since a rational expression equals 0 if and only if its numerator is 0, it follows that $(-22)^2 - 22 - 22b = 0$, or $22^2 + 22 - 22b = 0$. Dividing each side of this equation by 22 yields $22 + 1 - b = 0$, or $23 - b = 0$. Adding b to each side of this equation yields $23 = b$, so the value of b is 23.

Choice B is incorrect. This is the value of $g(0)$, not the value of b .

Choice C is incorrect. This is a value of x for which $g(x) = 0$, not the value of b .

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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